

Lab 6 - MITRE ATT&CK Investigation

Evidence-Based Technique Mapping Across SOC Scenarios

Attribute	Details
Framework	MITRE ATT&CK
Scenarios	Phishing, PowerShell, Defender detection, failed local logons
Objective	Map observed behavior without forcing unsupported techniques
Exercise type	Analyst interpretation lab

1. Objective

Review telemetry from prior labs and map only the behaviors directly supported by evidence. Automated SIEM/EDR ATT&CK labels were treated as hints rather than analyst conclusions.

2. Mapping Decisions

Scenario	Decision
Failed local logons	No ATT&CK technique assigned; evidence did not prove brute force or credential access.
Simulated phishing link	T1566.002 - Phishing: Spearphishing Link.
PowerShell activity	T1059.001 - Command and Scripting Interpreter: PowerShell.
Defender EICAR detection	No separate ATT&CK technique; this was defensive telemetry, not behavior by itself.

3. Evidence and Confidence

Behavior	Tactic / Technique	Confidence
Suspicious verification link and social engineering	Initial Access / T1566.002	High
powershell.exe directly observed	Execution / T1059.001	High
Defender Event ID 1116	No separate technique	High
4625 local failures	No technique - insufficient evidence	High

4. Analyst Principles

- Do not force mappings when evidence is insufficient.
- Map behavior, not severity.
- Validate automated ATT&CK tags against raw telemetry.
- Separate defensive telemetry from adversary behavior.
- State confidence and interpretation limits.

5. Sequence Interpretation

Phishing and PowerShell were reviewed as separate controlled scenarios. They should not be presented as a confirmed causal attack chain unless evidence directly links them.

6. Skills Demonstrated

- MITRE ATT&CK mapping
- Evidence validation
- SIEM/EDR rule interpretation
- Confidence-based analysis
- Avoiding over-attribution
- SOC reporting

7. Conclusion

The lab demonstrated disciplined ATT&CK mapping by assigning techniques only where the observed behavior supported them and leaving other events unmapped when the evidence was insufficient.